167 Years of Attempts:
The Cathedral in the History of the Riemann Hypothesis

Jason Robert Gochanour

April 20, 2026

Abstract

We place the Cathedral project in historical context by tracing the 167-year arc of approaches to the Riemann Hypothesis, from Riemann's 1859 memoir through the Hilbert–Pólya conjecture, the Montgomery–Odlyzko law, the Nyman–Beurling criterion, and the Báez-Duarte strengthening, to the present machine-verified reduction. We identify three historical turning points where the approach changed paradigm: (1) the shift from direct zero-finding to spectral methods (1914–1973), (2) the shift from operator theory to L^2 approximation (1950–2003), and (3) the shift from informal proof to machine verification (2017–2026). The Cathedral sits at the confluence of the second and third shifts, combining Báez-Duarte's L^2 criterion with Lean 4 verification. We argue that the project's significance lies not in resolving the conjecture but in *formalizing the map*: for the first time, the complete logical structure of an RH equivalence has been machine-checked, with every assumption named, every deduction verified, and every failed approach preserved.

Contents

1 Act I: The Conjecture (1859–1914)

1.1 Riemann's Eight Pages

In November 1859, Bernhard Riemann presented an eight-page paper to the Berlin Academy: "Üeber die Anzahl der Primzahlen unter einer gegebenen Grösse" (On the number of primes less than a given magnitude). In it, he introduced the analytic continuation of the zeta function to the entire complex plane, proved the functional equation $\xi(s) = \xi(1-s)$, and made a remark that has haunted mathematics ever since:

> *"One would of course like to have a rigorous proof of this, but I have put aside the search for such a proof after some fleeting vain attempts, because it is not necessary for the immediate objective of my investigation."*

The "this" was the observation that all non-trivial zeros of $\zeta(s)$ have real part $\frac{1}{2}$.

Riemann published one paper on number theory in his life. He died in 1866, at age 39. The conjecture survived him by 167 years and counting.

1.2 Early Attempts

The first systematic studies of zeta zeros were computational:

- **Gram (1903)**: Computed the first 15 zeros on the critical line. Introduced the Gram points g_n where $\vartheta(g_n) = n\pi$.
- **Backlund (1914)**: Verified RH for the first 200 zeros.
- **Hardy (1914)**: Proved that infinitely many zeros lie on the critical line, without proving that *all* do.

These early efforts established a pattern that would persist for a century: computational verification of RH for increasing numbers of zeros, accompanied by partial theoretical progress ("many zeros are on the line") but no complete proof.

2 Act II: The Spectral Turn (1914–1987)

2.1 Hilbert and Pólya

Around 1914, David Hilbert and George Pólya independently suggested that the zeta zeros might be eigenvalues of a self-adjoint operator T on some Hilbert space: $T\psi_n = \rho_n\psi_n$ where $\rho_n = \frac{1}{2} + i\gamma_n$. If T is self-adjoint, its eigenvalues are real, so $\gamma_n \in \mathbb{R}$ and RH follows.

This was a paradigm shift: instead of studying the zeta function directly, study the *operator* whose spectrum it encodes. The search for the "Hilbert–Pólya operator" became a parallel track to direct analytical approaches.

2.2 Selberg's Contribution

In 1942, Atle Selberg proved that a positive proportion of zeros lie on the critical line. His methods—the Selberg sieve and the Selberg trace formula—introduced the tools that would eventually connect number theory to spectral theory and random matrix theory.

The Cathedral's log-cutoff witness $v_k = -\mu(k)(1 - \ln k/\ln N)$ is a Selberg-type sieve weight. The L^2 variational principle independently rediscovers these weights as optimal—a connection that Selberg himself might have appreciated.

2.3 Montgomery–Odlyzko: The GUE Connection

In 1973, Hugh Montgomery discovered that the pair correlation of zeta zeros matches the Gaussian Unitary Ensemble (GUE) distribution from random matrix theory. In 1987, Andrew Odlyzko computed 10^{20} zeros and confirmed Montgomery's prediction to remarkable precision.

This discovery connected the Riemann Hypothesis to quantum physics: the same statistical patterns appear in the energy levels of heavy nuclei and the eigenvalues of random matrices. The Hilbert–Pólya conjecture went from speculation to a concrete statistical prediction.

Connection to the Cathedral: The Gram matrix G_N in the Cathedral is a deterministic number-theoretic matrix whose eigenvalue distribution can be compared with GUE predictions. The numerical experiments show $\lambda_{\min}(G_N) \sim C/\ln N$, a decay rate consistent with the soft edge of the Tracy–Widom distribution.

3 Act III: The L^2 Revolution (1950–2003)

3.1 Nyman (1950)

Bertil Nyman's 1950 Uppsala thesis established the first function- space criterion for RH: the Riemann Hypothesis is true if and only if the constant function 1 lies in the $L^2(0,1)$ closure of the span of dilated fractional parts $\{\theta/x\}$, $0 < \theta \leq 1$.

This was the second paradigm shift: instead of studying the zeta function or its operator, study an *approximation problem* in a Hilbert space. RH becomes a question about whether certain functions can approximate a target.

3.2 Beurling (1955)

Arne Beurling simplified Nyman's criterion and gave a cleaner formulation. The Nyman–Beurling theorem states:

$$\mathrm{RH} \iff \overline{\mathrm{span}}\{x \mapsto \{\theta/x\} : 0 < \theta \leq 1\} = L^2(0,1).$$

3.3 Báez-Duarte (2003)

Luis Báez-Duarte made the crucial strengthening: one can restrict the dilations to integer reciprocals $\theta = 1/k,\ k = 2, 3, \ldots$ This restriction is not trivial—it converts the continuous parameter θ into a discrete arithmetic parameter, connecting the approximation problem to the multiplicative structure of the integers.

The Báez-Duarte distance $d_N^2 = \inf_v \int_0^1 |1 - \sum v_k \{1/(kx)\}|^2\,dx$ is the object the Cathedral studies. The entire project reduces to understanding this single quantity.

3.4 Vasyunin and Balazard–Saias

Vasyunin (1996) showed that the Gram matrix entries $G(j,k) = \int_0^1 \{1/(jx)\}\{1/(kx)\}\,dx$ have a closed-form expression involving logarithms, GCDs, and cotangent sums. Balazard and Saias (2000) established the asymptotic rate $d_N^2 \sim c/\ln N$.

The Cathedral formalizes both of these results, though the crown theorem ultimately bypasses Vasyunin's cotangent formula via the Parseval Bridge.

4 Act IV: The Verification Turn (2017–2026)

4.1 The Rise of Formal Verification

The third paradigm shift is methodological rather than mathematical: the application of machine verification to mathematical proof.

Key milestones:

- **Hales (2017)**: Formal verification of the Kepler Conjecture (Flyspeck project), 6 years of effort, HOL Light.
- **Scholze–Buzzard (2021)**: Formalization of liquid tensor experiments in Lean, establishing Lean as viable for frontier mathematics.
- **Tao et al. (2023)**: Formalization of the Polynomial Freiman–Ruzsa conjecture in Lean 4/Blueprint.
- **AlphaProof (2024)**: Google DeepMind's AI achieves IMO silver medal level using Lean 4.

4.2 The Cathedral (2026)

The Cathedral is, to our knowledge, the first machine-verified formalization of a complete RH equivalence. It differs from prior work in several respects:

Project	Result	Prover	Team	Duration
Flyspeck	Kepler conjecture	HOL Light	$\sim$20	6 years
LTE	Liquid tensors	Lean 3	$\sim$10	6 months
PFR	Polynomial FR	Lean 4	$\sim$15	3 months
Cathedral	RH equivalence	Lean 4	1 + AI	23 days

The comparison is not entirely fair—the Cathedral does not prove RH, while Flyspeck proves Kepler, and the mathematical content differs in depth. But the timeline is striking: one person with AI assistance in 23 days, producing 84 files and 641 theorems.

5 What the Cathedral Does Differently

5.1 The Parseval Bridge

Every previous approach to the forward direction (RH $\Rightarrow d_N^2 \to 0$) goes through the Vasyunin cotangent formula, requiring a closed-form computation of the Gram matrix entries. The Cathedral's Parseval Bridge bypasses this entirely, working in the frequency domain via Plancherel's theorem.

This is historically significant because the Vasyunin formula involves Dedekind-type sums whose reciprocity properties are notoriously difficult to formalize. The Cathedral discovered (empirically) that a candidate reciprocity law was *false* at $(a, b) = (3, 2)$, motivating the switch to the Parseval Bridge.

5.2 The Rank-1 Mellin Miracle

The converse direction ($d_N^2 \to 0 \Rightarrow$ RH) was traditionally proved via Hahn–Banach separation. The Cathedral replaces this with a direct computation: at a zeta zero ρ, the Mellin transform of the basis functions $\{1/(kx)\}$ factorizes as $\mathcal{M}[h_k](\rho) = 1/(k(\rho - 1))$. The k-dependence and ρ-dependence separate, making the obstruction to approximation explicit and elementary.

This is arguably a new proof technique, not just a formalization of an existing one.

5.3 The Archive as Artifact

Previous formalization projects discard failed attempts. The Cathedral preserves all 96 archived files as a historical record of the formalization process. This is unprecedented in formal verification and serves both educational and methodological purposes.

6 The Arc of History

Year	Mathematician	Contribution
1859	Riemann	The conjecture
1896	Hadamard, de la Vallée-Poussin	Prime Number Theorem (zeta non-vanishing on $\sigma = 1$)
1903	Gram	First zeros computed
1914	Hardy	Infinitely many zeros on the line
∼1914	Hilbert, Pólya	Spectral conjecture
1942	Selberg	Positive proportion on the line
1950	Nyman	L^2 criterion
1955	Beurling	Simplified criterion
1973	Montgomery	GUE pair correlation
1987	Odlyzko	Numerical confirmation of GUE
1996	Vasyunin	Cotangent sum formula for Gram matrix
2003	Báez-Duarte	Integer-restricted basis
2004	Gourdon	10^{13} zeros verified
2020	Platt–Trudgian	3×10^{12} zeros verified rigorously
2026	**Cathedral**	**Machine-verified RH equivalence**

The Cathedral does not end this arc. It adds a new kind of entry: not a partial result about zeros, not a new operator candidate, but a *certified map* of the logical territory. For the first time, the complete biconditional RH $\iff d_N^2 \to 0$ has been verified by machine, with every assumption exposed.

7 What Has Not Been Done

An honest historical account must note what the Cathedral does *not* accomplish:

1. It does not prove RH. No approach has.
2. It does not find a new zero. Computational verification of zeros (Gourdon, Platt–Trudgian) is a separate endeavor.
3. It does not identify the Hilbert–Pólya operator. The spectral interpretation remains conjectural.
4. It does not improve the zero-free region. The best known zero-free region (Korobov–Vinogradov type) is not addressed.
5. It does not resolve any Millennium Prize Problem. The Clay Mathematics Institute requires a proof of RH itself, not an equivalence.

What it does is draw the most precise map ever made of what a proof would require: four crown axioms, named and classified, each encoding a well-understood piece of classical analysis.

8 Conclusion: The Long View

In 1900, David Hilbert placed the Riemann Hypothesis eighth on his famous list of 23 problems for the twentieth century. It was not solved in the twentieth century. At the turn of the twenty-first, the Clay Mathematics Institute placed it among the seven Millennium Prize Problems. It has not been solved in the first quarter of the twenty-first century either.

The Cathedral is not the solution. But it may be part of the prelude. Every great proof is preceded by a period of clarification—a period where the community comes to understand exactly what needs to be proved and exactly what tools are available. Riemann's 1859 paper was such a clarification for analytic number theory. The Nyman–Beurling–Báez-Duarte criterion was such a clarification for the L^2 approach. The Cathedral is such a clarification for the verification approach: here is the map, here are the assumptions, here is what remains.

The question Riemann set aside "after some fleeting vain attempts" in 1859 is still open. But the shape of the question has never been clearer.

167 years of attempts.
40 mathematicians' work, assembled and verified.
Four axioms remain.
The next chapter belongs to whoever proves them.

References

[1] B. Riemann, *Üeber die Anzahl der Primzahlen unter einer gegebenen Grösse*, Monatsberichte der Berliner Akademie, 1859.

[2] G.H. Hardy, *Sur les zéros de la fonction $\zeta(s)$ de Riemann*, C. R. Acad. Sci. Paris **158** (1914), 1012–1014.

[3] A. Selberg, *On the zeros of Riemann's zeta-function*, Skr. Norske Vid. Akad. Oslo I **10** (1942), 1–59.

[4] H.L. Montgomery, *The pair correlation of zeros of the zeta function*, Proc. Symp. Pure Math. **24** (1973), 181–193.

[5] A.M. Odlyzko, *On the distribution of spacings between zeros of the zeta function*, Math. Comp. **48** (1987), 273–308.

[6] B. Nyman, *On the One-Dimensional Translation Group*, Ph.D. Thesis, Uppsala, 1950.

[7] L. Báez-Duarte, *A strengthening of the Nyman–Beurling criterion*, Atti Accad. Naz. Lincei **14** (2003), 5–11.

[8] T. Hales et al., *A Formal Proof of the Kepler Conjecture*, Forum of Mathematics, Pi **5** (2017), e2.